

|                |                                                                                     |
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| <b>EX NO:1</b> | <b>Butterworth IIR Filters</b>                                                      |
| <b>DATE:</b>   | <b>1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.</b> |

**Code:**

```

clc;
clear;
close all;

% EX NO: 2
% Chebyshev IIR Filter
% 2.1 Design and Performance Evaluation of a Chebyshev Type-I
% Low-Pass Filter using MATLAB
% Filter Specifications
Wp = 0.40; % Passband Edge Frequency (Normalized)
Ws = 0.55; % Stopband Edge Frequency (Normalized)
Rp = 1; % Passband Ripple (dB)
As = 40; % Stopband Attenuation (dB)
% Calculate Minimum Filter Order
[N, Wn] = cheblord(Wp, Ws, Rp, As);

% Design Chebyshev Type-I Low-Pass Filter
[b, a] = cheby1(N, Rp, Wn);

% Display Results
fprintf('Filter Order = %d\n', N);
fprintf('Cutoff Frequency = %.2f x pi rad/sample\n', Wn);

% Stability Check
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% Frequency Response
figure;
freqz(b, a);
title('Frequency Response');

% Impulse Response
figure;
impz(b, a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');

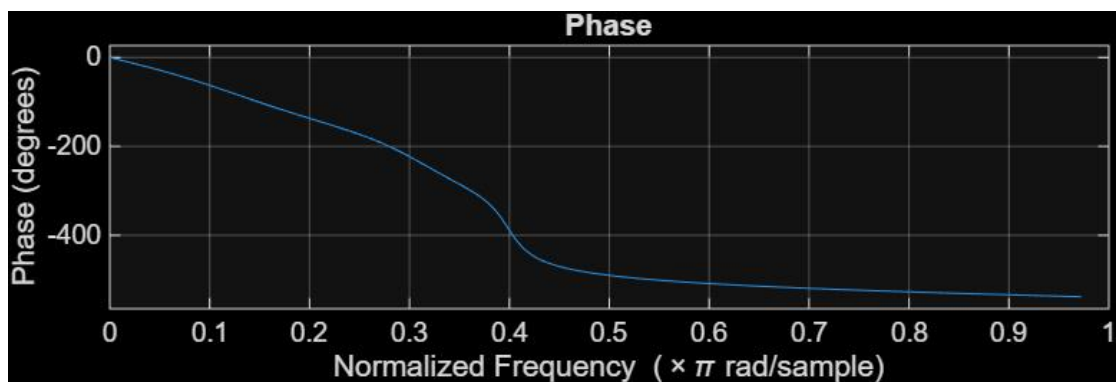
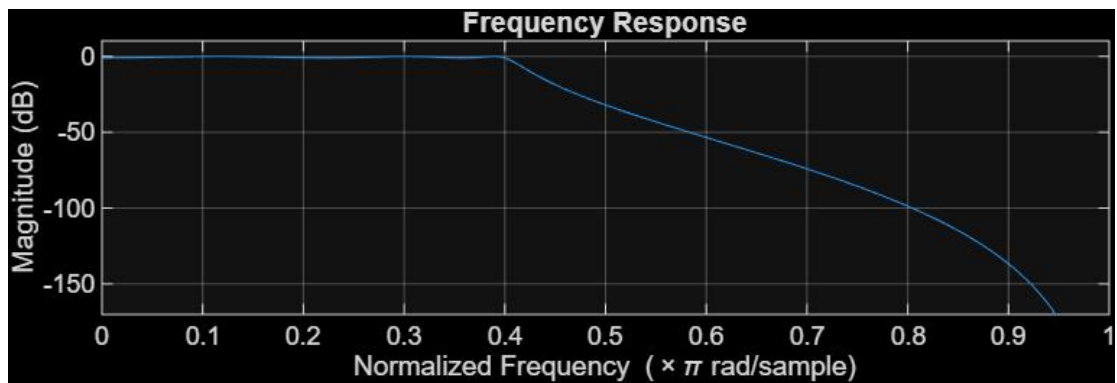
```

# Outputs:

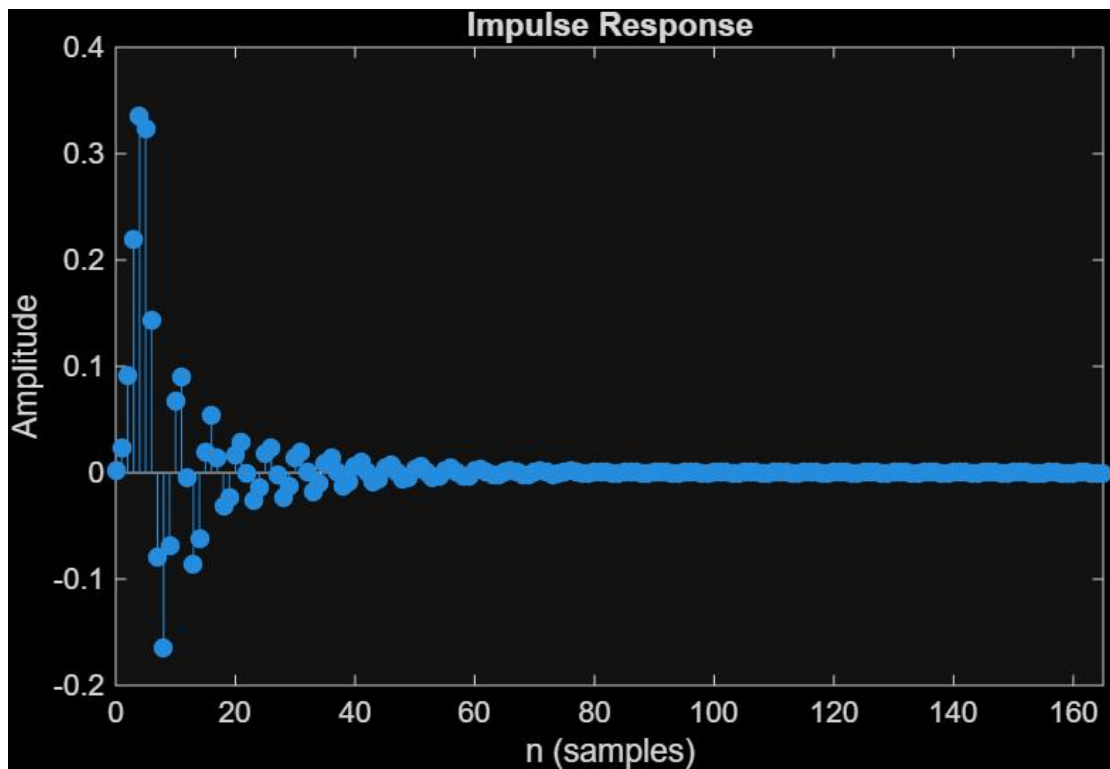
## Objective 1:

```
Command Window  
Filter Order = 6  
Cutoff Frequency = 0.40 × pi rad/sample  
System is STABLE  
>>
```

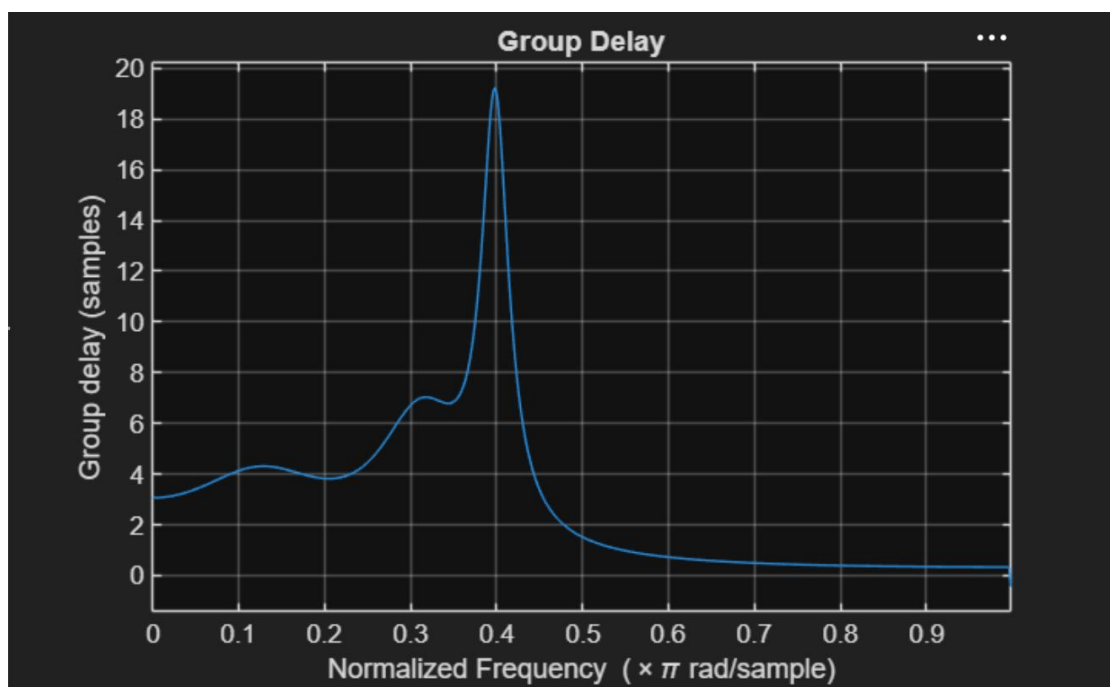
## Objective 2:



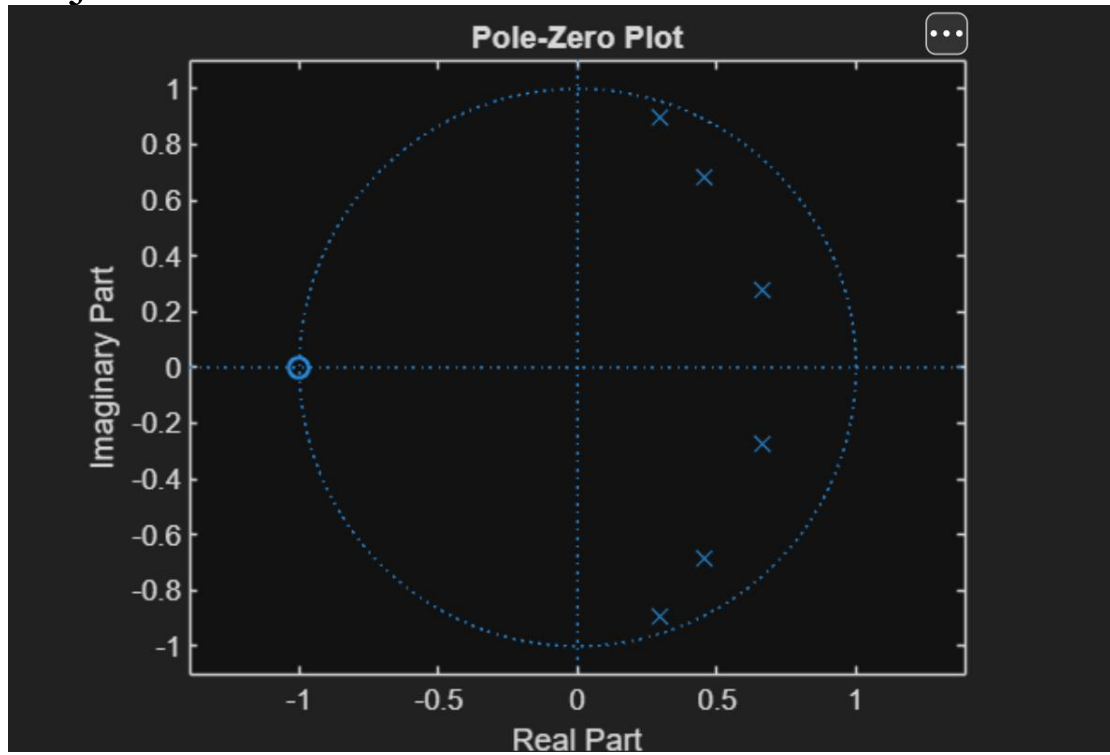
Objective 3:



Objective 4:



## Objective 5:



|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| <b>EX NO:2</b> | <b>Chebyshev IIR Filter</b>                                                              |
| <b>DATE:</b>   | <b>2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter</b> |

## CODE :

```

clc;
clear;
close all;

% EX NO: 2
% Chebyshev IIR Filter
% 2.2 Design and Performance Evaluation of a
% Chebyshev Type-I IIR High-Pass Filter Using MATLAB

% Filter Specifications
Rp = 1;    % Passband Ripple (dB)
Rs = 40;   % Stopband Attenuation (dB)
Wp = 0.50; % Passband Edge Frequency (Normalized)
Ws = 0.35; % Stopband Edge Frequency (Normalized)

% Calculate Minimum Filter Order
[n, Wn] = cheb1ord(Wp, Ws, Rp, Rs);

% Design Chebyshev Type-I High-Pass Filter
[b, a] = cheby1(n, Rp, Wn, 'high');

% Frequency Response
figure;
freqz(b, a);
title('Frequency Response');

% Impulse Response
figure;
impz(b, a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);

```

```

title('Pole-Zero Plot');

% Stability Check
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% Display Filter Parameters
fprintf('Filter Order = %d\n', n);
fprintf('Cutoff Frequency = %.4f\n', Wn);

```

## OUTPUTS :

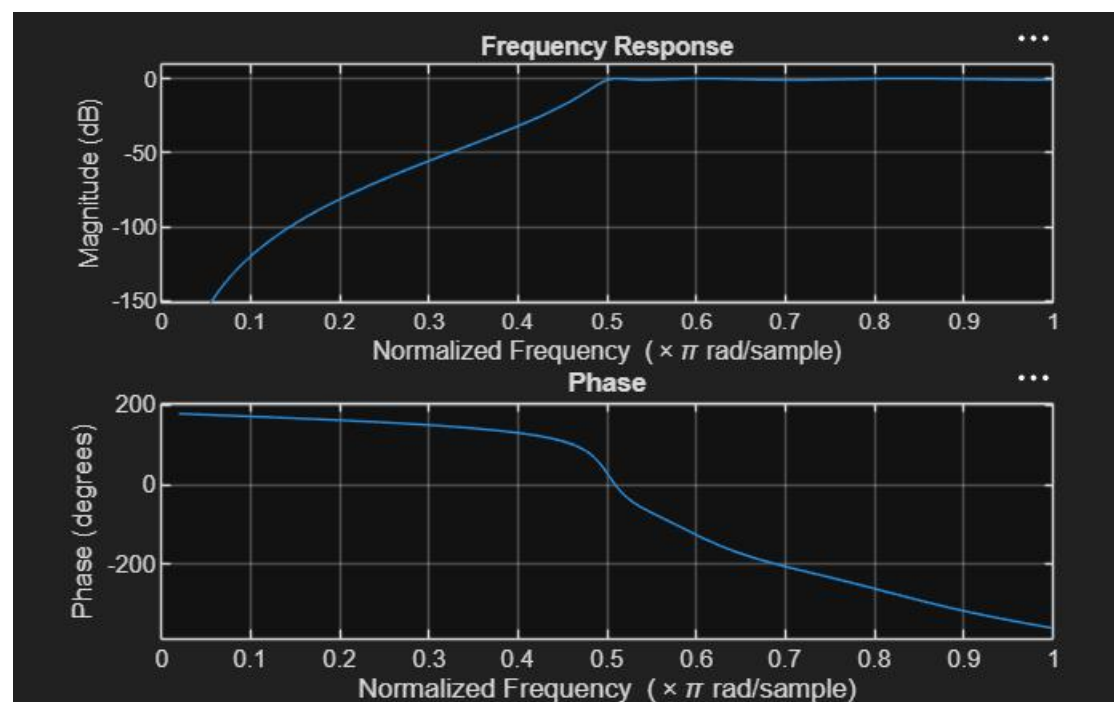
### Objective 1:

```

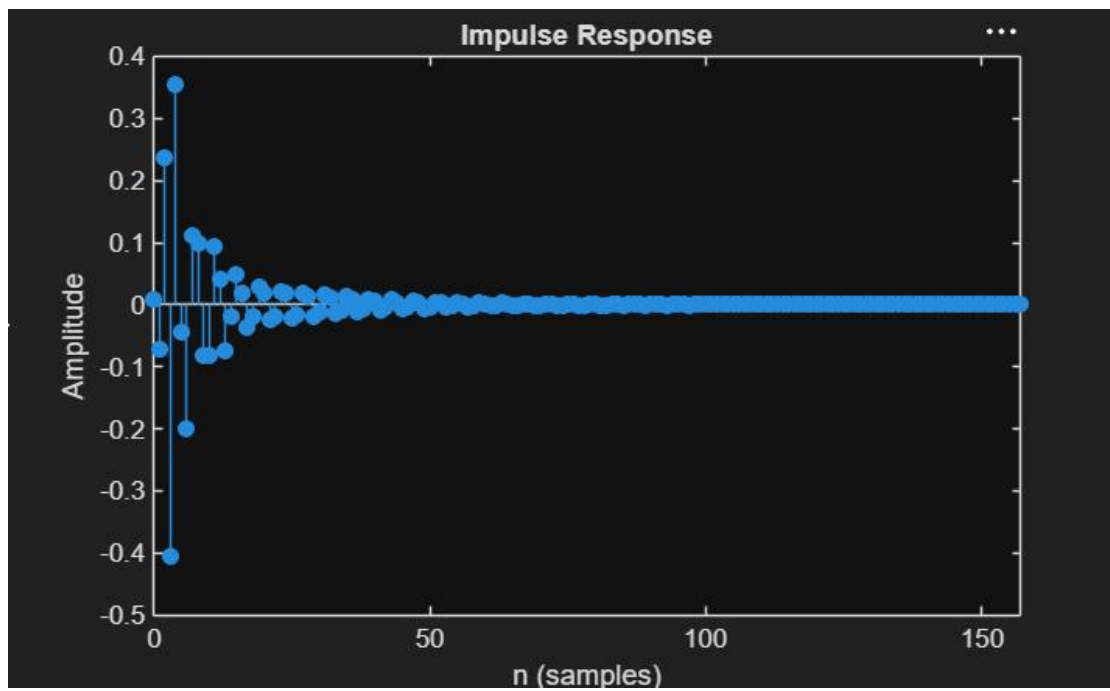
Command Window
System is STABLE
Filter Order = 6
Cutoff Frequency = 0.5000
>>

```

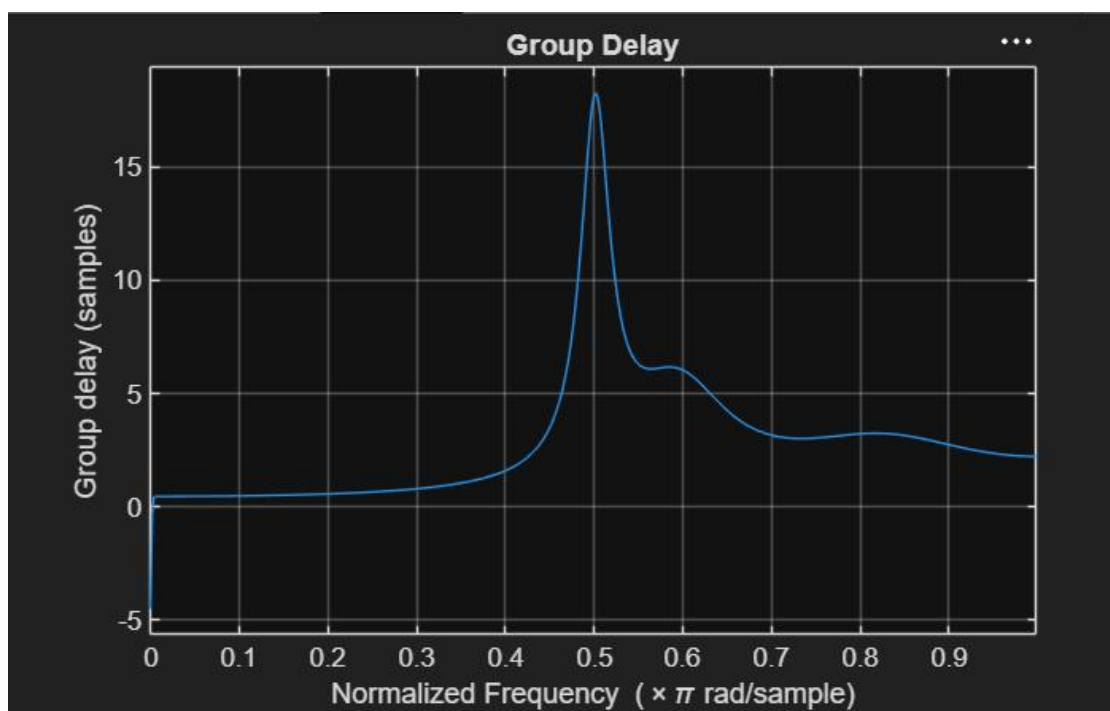
### Objective 2:



### Objective 3:



### Objective 4:



## Objective 5:

